requirements | Status: mixed | App v1.0.20 | Reviewed 2026-10-24

Requirements

This repo has no formal requirements document. The requirements below were **reconstructed** from the README and from behaviour implemented in code **[inferred]**. Each requirement has a stable id so tests and issues can refer to it. The **Evidence** column points to the implementation.

1. Goals

Id	Goal
G-1	Show state-of-the-art open models running **fully on-device** on Android.
G-2	Let developers and enthusiasts evaluate models on their own hardware (quality, latency, memory).
G-3	Showcase agentic patterns on-device: skills, function calling, MCP, device intents.
G-4	Act as a reference implementation of LiteRT-LM, AICore, and MediaPipe integration.

2. Non-goals

Id	Non-goal
NG-1	Cloud inference or server-side processing of user content.
NG-2	Being a general-purpose production assistant. The app is labelled an experimental Beta.
NG-3	Accepting external code contributions (for now; see `CONTRIBUTING.md`).

3. Functional requirements

Id	Requirement	Evidence
FR-01	Show a home screen of tasks grouped by category, built from the registered tasks.	`HomeScreen`, `Set<CustomTask>`
FR-02	Load the model catalog for the running app version from a remote allowlist, with a local cache fallback.	`loadModelAllowlist()`
FR-03	Hide models the device can't run (disabled, unsupported AICore device, NPU-only on an unsupported SoC).	`loadModelAllowlist()` filters
FR-04	Download models in the background with progress, resume after interruption, and allow cancel and delete.	`DownloadWorker`, `DownloadRepository`

Id	Requirement	Evidence
FR-05	Support gated Hugging Face models through OAuth sign-in.	`ModelManagerViewModel.handleAuthResult`
FR-06	Let users import their own `.litertlm` / `.task` models and test them in chat.	`ModelImportDialog`, `llm_test`
FR-07	Multi-turn chat with streaming, optional thinking output, image and audio input when the model supports them.	`LlmChatScreen`, `LlmChatModelHelper`
FR-08	Ask Image: up to 10 images per session (1 on AICore).	`MAX_IMAGE_COUNT`, `MAX_IMAGE_COUNT_AI_CORE`
FR-09	Audio Scribe: one clip of up to 30 s at 16 kHz.	`MAX_AUDIO_CLIP_*`, `SAMPLE_RATE`
FR-10	Prompt Lab: single-turn templates with editable sampling parameters.	`ui/llmsingleturn/*`
FR-11	Agent Skills: load skills on demand; run JS skills, Android intents, and MCP tools; render image and webview results.	`AgentChatTask`, `tools/*`
FR-12	Users can add skills from a URL or a local folder, enable or disable them, edit them, and test them.	`SkillManager`, `SkillManagerBottomSheet`
FR-13	Users can add MCP servers (StreamableHTTP, optional header auth) and control per-tool permission.	`McpManagerViewModel`, `RunMcpTool`
FR-14	Ask the user for consent before an MCP tool call unless that tool is set to always-allow.	`AskMcpToolCallPermissionAction`
FR-15	Natural-language and image-to-image search over the user's photos or a sample album.	`customtasks/smartalbum/*`
FR-16	Find moments in videos by description; trim, save, and export clips.	`customtasks/videomomentfinder/*`
FR-17	Segment objects in photos and build collages.	`customtasks/scrapbook/*`
FR-18	Offline device actions and a mini-game driven by function calling.	`mobileactions/*`, `tinygarden/*`
FR-19	Persist chat history per task and let users resume or delete sessions.	`LlmSessionManager`, `ChatSessionRepository`
FR-20	Benchmark LiteRT-LM models and keep the results.	`ui/benchmark/*`
FR-21	Deep links open a task or model; an external query must be confirmed by the user before it is sent.	`GalleryNavGraph`, `ExternalPromptConfirmationDialog`
FR-22	The user can opt out of analytics.	`Settings.disable_firebase_analytics`

4. Non-functional requirements

Id	Category	Requirement	Status / evidence
NFR-01	Privacy	Prompts, images, audio, and model outputs are processed on-device only.	**[current]** No inference network calls. Remote MCP and skills are user-opt-in.
NFR-02	Offline	Every task works offline once its model is downloaded. MCP and remote skills are the exceptions.	**[current]**
NFR-03	Platform	Android 12+ (API 31). Targets API 37.	`minSdk 31`, `targetSdk 37`
NFR-04	Device fit	Warn when RAM is below a model's `minDeviceMemoryInGb` (6–12 GB for current models).	`MemoryWarning`
NFR-05	Performance	Choose the best available accelerator (GPU/NPU/TPU) with a CPU fallback. Record TTFT, prefill/decode speed, and memory.	`BackendSpec`, `common/metrics/*`
NFR-06	Reliability	Downloads survive app kill (WorkManager foreground) and resume over HTTP Range.	`DownloadWorker`
NFR-07	Reliability	The catalog loads from cache if GitHub is unreachable.	`readModelAllowlistFromDisk`
NFR-08	Security	Untrusted inputs (deep-link queries, MCP calls, skill installs) need explicit user consent.	Disclaimers and permission dialogs
NFR-09	Security	Credentials (HF tokens, skill secrets, MCP headers) are stored only on the device.	**[current]** Stored, but **unencrypted**. See 10-security-privacy.
NFR-10	Accessibility	Content descriptions on icons, semantic headings, and support for dynamic text.	Partial **[inferred]**: `cd_*` strings and `semantics { heading() }` exist. No audit found.
NFR-11	Localization	All user-facing strings in resources with translator notes.	Mostly. About 17 files have hard-coded text.
NFR-12	Maintainability	Features plug in without editing central code (Hilt `@IntoSet`).	**[current]**
NFR-13	Quality	Automated tests gate merges.	**Not met.** No tests exist (see 06-testing). **[proposed]**
NFR-14	Observability	Opt-in, anonymous usage and performance telemetry.	Firebase Analytics (absent in OSS builds)